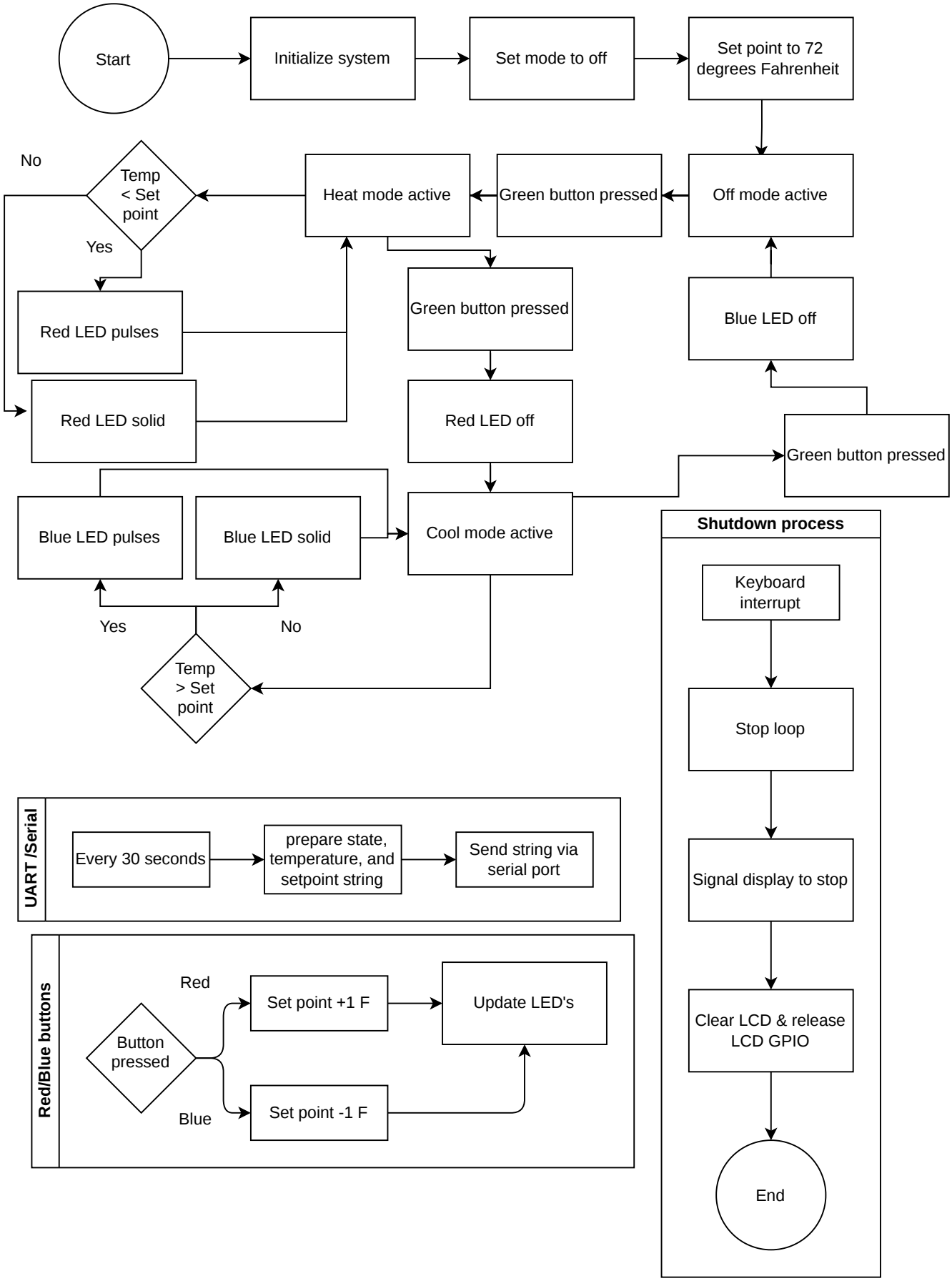
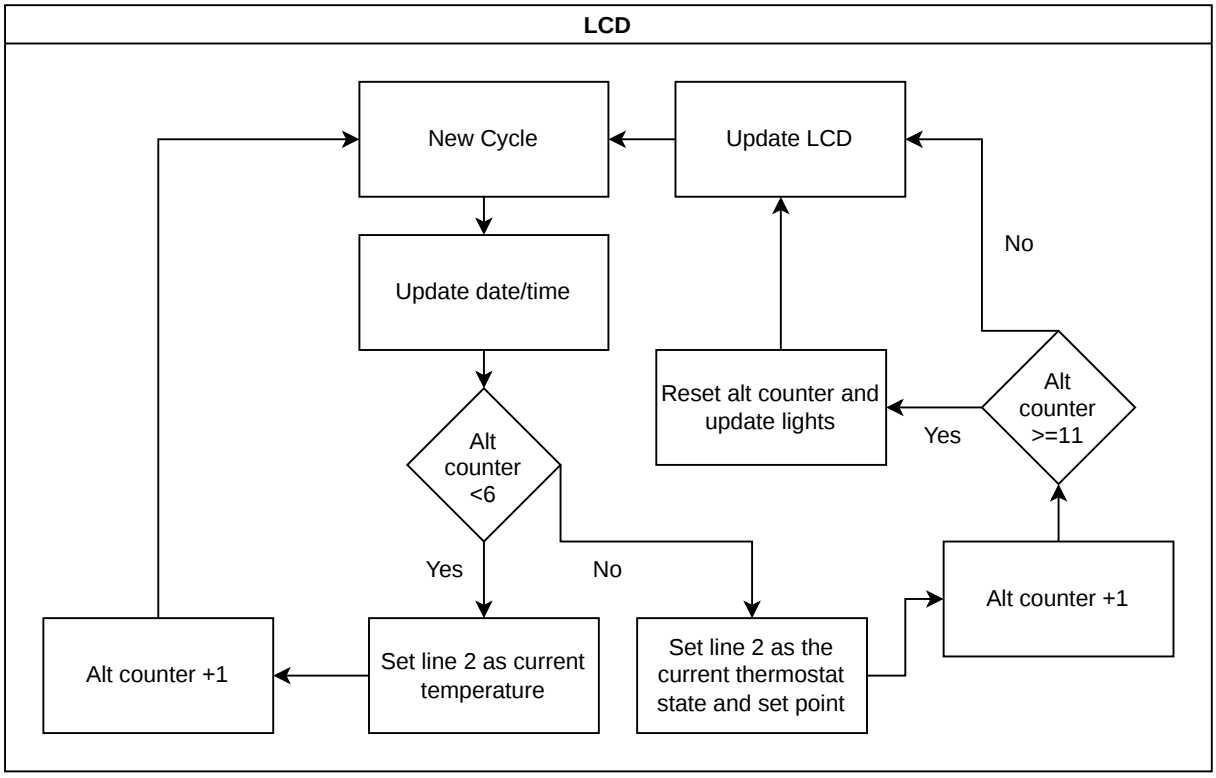

Start
Initialize system
Set mode to off
Set point to 72 degrees Fahrenheit
Off mode active
Green button pressed
Heat mode active
Temp < Set point
No
Yes
Red LED pulses
Red LED solid
Green button pressed
Red LED off
Cool mode active
Blue LED pulses
Blue LED solid
Yes
No
Temp > Set point
Blue LED off
Green button pressed
Shutdown process
Keyboard interrupt
Stop loop
Signal display to stop
Clear LCD & release LCD GPIO
End
UART /Serial
Every 30 seconds
prepare state, temperature, and setpoint string
Send string via serial port
Red/Blue buttons
Button pressed
Red
Blue
Set point +1 F
Set point -1 F
Update LED's


LCD
New Cycle
Update LCD
Update date/time
Reset alt counter and update lights
Alt counter >=11
No
Yes
Alt counter <6
Yes
No
Alt counter +1
Set line 2 as current temperature
Alt counter +1
Set line 2 as the current thermostat state and set point